

Data Cleaning and Quality Assessment

Homework 3 – Repurposing Data Sources

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1. Extracting information from HTML documents

I downloaded both Wikipedia articles using 'requests' and parsed them with BeautifulSoup. The vaccination-related tables were extracted and saved as CSV files. HTML was convenient because tables have clear rows and columns. The U.S. and Australian tables contain similar vaccination information, but their columns and reporting dates are not exactly the same.

2. Extracting information from PDF documents

The HTML extraction produced cleaner tables because HTML explicitly stores rows and cells. The PDF extraction preserved much of the visible information, especially when pdftotext -layout was used, but some rows, columns, footnotes, and spacing required additional cleaning.

3. Analysis questions

3a. Which format was easier to parse?

HTML was easier to parse.

HTML tables already have rows and columns, so BeautifulSoup can extract the cells directly. PDF files mainly preserve the appearance of the document, so additional processing is needed.

3b. Which format preserved tables more accurately?

HTML preserved the table structure more accurately.

The table rows and columns remained clearly identifiable. PDF extraction preserved the general layout, but some column boundaries and row relationships could be distorted.

3c. What information was lost or distorted during PDF extraction?

Some information could become difficult to interpret after PDF extraction. Examples include wrapped text, footnote symbols, repeated table headers, irregular spaces, and values becoming separated from their original row or column. Images and icons also may not be extracted as useful table information.

3d. What preprocessing steps improved extraction quality?

The most helpful steps were using pdftotext -layout, removing blank lines, identifying the correct table section, splitting columns using repeated whitespace, cleaning footnote symbols, converting numeric values to numbers, and checking several extracted rows against the original PDF.

3e. Which format would you recommend for large-scale data collection and why?

I would recommend HTML for large-scale data collection.

HTML is easier to automate because tables have a structured format. It also requires less cleaning and makes it easier to validate rows and columns. PDF is useful when an HTML version is not available, but it usually requires more preprocessing.

Extra credit: Visualization and insight generation

Why did I select these visualizations?

I selected a coverage chart and a timeline because they provide two simple ways to understand the vaccination rollout. The first focuses on vaccination percentages, while the second focuses on when important rollout events occurred.

Key insights:

The two countries reported vaccination information in somewhat different ways. The U.S. article provides detailed state and territory information, while the Australian article also describes rollout phases and vaccination progress. The timeline makes the different starting points and milestones easier to see.

Which country's information was easier to interpret?

The Australian rollout information was relatively easy to interpret because the article presents several rollout phases and milestones clearly. However, for machine extraction, HTML was easier for both countries because the table structure was preserved.